



Spring 2026

Robots Motion Planning and Control Course 362-2-5481

Mini-Project

General

1. The work should be submitted to Moodle in a word or PDF file.
Make sure to include in the PDF a **link to a Google Drive folder** that contains: a PDF file of the paper that you selected in part 2 of the mini-project, PPTX file of the presentation, and program files, videos, and other related materials.
Make sure to share the link with amir.shapiro@gmail.com, and have the required read permissions.
2. The work should be done by groups of 1,2, or 3 students. Only one student from the group should submit the work. Please clearly state on the title page who are the group members.
3. The mini-project is **due before June 18**.
4. People that served in reserve duty more than a 10 days during the semester can select to do only part 1 or 2 (do not need to do both)

Part 1

1. Choose a 3 or more DOF (degrees of freedom) robot arm.
2. Ask the lecturer approval to your dynamical system by email **before May 7, 2026**
3. Develop the dynamical equations of motion (at least 3, time dependent, differential equations)
4. Develop a simulation with numerical integration of the equations.
5. Choose start position with zero velocity.
6. Provide motion graphs of the configuration parameters with respect to time, and a “stick-figure” when the inputs are zero.
7. Chose target position for the system.
8. Design a PID controller.
9. Provide motion graphs of the configuration parameters with respect to time, and a “stick-figure” when the inputs are determined by the PID controller.

Note: there is **part 2** in the next page.



Part 2

1. Choose a research paper that deals with motion planning and/or control of robots. The paper should be on a new method that was not taught in class. The paper should be from a leading journal or conference in robotics (from the past 5 years), examples are:
 - a. Science Robotics
 - b. IEEE Transactions on Robotics (TRO)
 - c. International Journal on Robotics Research (IJRR)
 - d. IEEE Robotics and Automation Letters (RA-L)
 - e. IEEE conference on Robotics and Automation (ICRA)
 - f. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

Note 1: The work can be your own original research as well (and not necessarily a published paper).

Note 2: You can select a paper that deals with Machine Learning, AI and Robotics.

2. Ask the lecturer approval of your paper (or work subject) by email **before May 27, 2024**
3. Register your group for presentation of your work in class, in the following file:
<https://docs.google.com/spreadsheets/d/1zugIsUnbiVW1IKj4TyyI172CTSLDz1v3YKAFJ8xLH4g/edit?usp=sharing>
4. Write a brief description of the paper's contributions.
5. Implement the method that was presented in the paper on a simulation software:
 - a. Model the robot and the control or motion planning algorithms in a programming language of your choice (Matlab, Python, ROS, etc.)
 - b. Implement the motion planning or control algorithm.
 - c. Run the simulation and generate graphs of the robot configuration, both in configuration space, and in task space. And other graphs and videos that show and explain the solution.
 - d. Record a video of your screen showing the robot in motion.
6. Present the work in class to teach all student the new method.



- a. The presentation will be done **in class on July 1.**
- b. In the presentation show the main contributions of the paper, and your implementation, including program structure, main functions, run examples, results, and conclusions.
7. Submit a written report
 - a. The report is due **before June 18.**
 - b. Title page with students' names and ID numbers.
 - c. The paper that your work is about. Include a copy.
 - d. Main contributions of the paper/work.
 - e. Description of the algorithm and explanation of how it works.
 - f. The simulation program structure.
 - g. Main functions of the simulation and explanation of how they work.
 - h. Simulation run example
 - i. The problem used in the simulations
 - ii. Results of the simulations in graphs, and figures.
 - iii. A link to a video of the simulation
 - iv. A link to the program files, and instructions on how to install and run them.
 - i. Conclusions
 - i. What has been done in the work?
 - ii. What are the results and insight that you have achieved in the work?
 - iii. What are the advantages and disadvantages of using the algorithm/method?
 - iv. Recommendation for improving and extending the algorithm/method.